

Robert Burdsall

704-305-0517 | robertaburdsall@gmail.com | [linkedin.com/in/robertburdsall](https://www.linkedin.com/in/robertburdsall) | github.com/robertburdsall

OBJECTIVE

Electrical and Computer Engineering student at NC State with industry experience in PLC programming, electrical panel design, and machine vision integration, seeking a Summer 2027 controls or automation engineering internship to apply industrial automation expertise in manufacturing systems.

EDUCATION

North Carolina State University

Bachelor of Engineering in Electrical and Computer Engineering. GPA: 3.5

Raleigh, NC

Aug. 2025 – May 2028

North Carolina School of Science and Mathematics

High School Diploma, Computer Science Focus. GPA: 4.8

Morganton, NC

Aug. 2023 – May 2025

EXPERIENCE

IDEAL Fastener Corporation

Electrical Engineering Intern

May 2026 – Aug. 2026

Oxford, NC

- Designed and fabricated 5+ electrical control panels for production machinery, handling full wiring assembly (relays, power supplies, DIN rail, disconnects, proximity/laser sensors, solenoid valves) and producing TinyCAD schematics for future replication.
- Programmed Omron PLCs using CX-Programmer to implement automated machine control logic, and configured and trained Keyence IV3/IV4 vision systems — defining inspection criteria and teaching defect recognition — to upgrade 3 existing floor machines for real-time quality detection.
- Led end-to-end design of a zipper cap inspection machine inspecting 120 caps/min, featuring trained Keyence vision inspection, vacuum ejection, solenoid valve sequencing, a DOPSoft HMI, and PLC-driven quality tracking with alarm functionality for good/bad rate monitoring.

FIRST Robotics Competition

North Carolina School of Science and Mathematics, Team 9150

Jan. 2024 – Present

Morganton, NC

- Designed and implemented full robot control software in Java using WPILib, achieving 3rd-highest scoring performance in North Carolina with integrated vision and dual scoring systems.
- Lead Mentor for 25+ students, teaching Java, WPILib, and robotics programming fundamentals.

Computer Science Teaching Assistant

North Carolina School of Science and Mathematics

Aug. 2024 – May 2025

Morganton, NC

- Taught 150+ students programming fundamentals in Java and Python.
- Built a cross-platform resource tracking app using Flutter/Dart with faculty.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, Dart, SQL, JavaScript, HTML/CSS

Developer Tools: Git, Docker, XCode, Android Studio, Google Cloud Platform, VS Code, TensorFlow, IntelliJ

Design: Solidworks, Prusa Slicer, Cura, TinyCAD, Keyence SmartNavigator, CXProgrammer, DOPSoft

VOLUNTEERING

Boy Scouts of America

Eagle Scout, Troop 222. Senior Patrol Leader | Assistant Scoutmaster

2015 – Present

Huntersville, NC

- Assistant Scoutmaster mentoring 30+ Scouts and leading outdoor programs to build leadership, teamwork, and problem-solving skills.

PROJECTS

Aurum

Dart, Python, Flutter, ExpressJS, TensorFlow

Sep. 2024 – Dec. 2024

Morganton, NC

- Developed a cross-platform Flutter mobile app for real-time coin appraisal, leveraging TensorFlow Lite.
- Engineered live coin value retrieval and portfolio tracking features, integrating both client-side and server-side logic for seamless data synchronization.
- Implemented a robust backend infrastructure using Express.js, JSON APIs, and JWT-based authentication.